

Homework Assignment 10
Econ 629 - Texas A&M University
Fall 2012

Posted: Thursday, Nov. 8th, 2012.

Due: Tuesday, Nov. 13th (Class time).

1. Problems 1, 3, 4 in Section 6.7 Lecture notes.
2. Consider an expected utility maximizer with utility index $u(x) = -e^{-ax}$ with $a > 0$ and initial wealth $w > 0$. There is a risk free asset that has a return of 2 (if the agent invest x dollars, then she receives $2x$). There is also a risky asset with random return 1 with probability $\alpha > 0$ and 3 with probability $(1 - \alpha)$.
 - (a) Set up the portfolio choice problem for the agent (denote the amount invested in the risky asset by x and its optimal value by x^*).
 - (b) Under what conditions is $x^* > 0$? Interpret.
 - (c) Under what conditions is $x^* < w$? Interpret.
 - (d) Assume that the conditions guaranteeing that x^* is interior are satisfied. Calculate x^* .
 - (e) How does x^* depend on a ? Interpret.